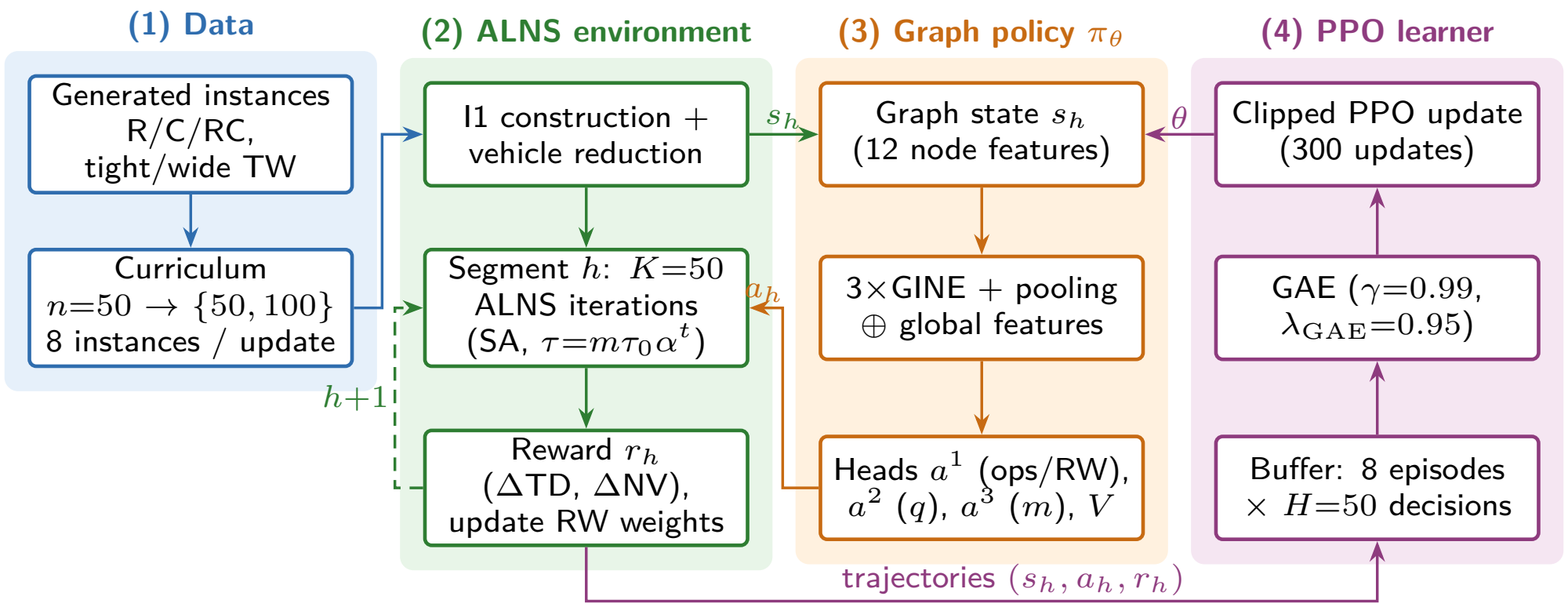

(1) Data
Generated instances
R/C/RC,
tight/wide TW
Curriculum
$n{=}50 \rightarrow \{50, 100\}$
8 instances / update
(2) ALNS environment
I1 construction +
vehicle reduction
Segment $h$: $K{=}50$
ALNS iterations
(SA, $\tau{=}m\tau_0\alpha^t$)
Reward $r_h$
($\Delta$TD, $\Delta$NV),
update RW weights
$h{+}1$
$s_h$
$a_h$
(3) Graph policy $\pi_\theta$
Graph state $s_h$
(12 node features)
3×GINE + pooling
$\oplus$ global features
Heads $a^1$ (ops/RW),
$a^2$ ($q$), $a^3$ ($m$), $V$
$\theta$
(4) PPO learner
Clipped PPO update
(300 updates)
GAE ($\gamma{=}0.99$,
$\lambda_{\mathrm{GAE}}{=}0.95$)
Buffer: 8 episodes
× $H{=}50$ decisions
trajectories $(s_h, a_h, r_h)$